Lightweight Spectral–Spatial Attention Network for Hyperspectral Image Classification

Ying Cui, Jinbiao Xia, Zhiteng Wang, Shan Gao, and Liguo Wang, *Member, IEEE*

***Abstract*— Convolutional neural networks (CNNs) have exhibited extraordinary achievements in hyperspectral image (HSI) classification due to their detailed representation of features. However, the improvement of classification accuracy often leads to an evident increase in the complexity of the model, which makes it challenging for the model with the state-of-the-art performance to be applied in the actual scene. Considering MobileNetV3 as a lightweight feature extractor, this article proposes a model suitable for HSI classification based on MobileNetV3. To decrease the problem of massive redundant calculations in the existing spatial attention module, this article proposes a more concise and efficient spatial attention module based on the visual feature maps experiment. Besides, multiclass focal-loss is applied to solve the problem that the difficulty of classification varies for each sample. The experimental results demonstrate that in the case of using very few training sets, the proposed model can tremendously reduce the number of calculations and parameters while maintaining high accuracy.**

***Index Terms*— Attention mechanism, feature maps visualization, lightweight model, multiclass focal-loss.**

I. Introduction

AT PRESENT, remote-sensing image has been widely used in contemporary society [1], [2], and the emergence of hyperspectral image (HSI) is considered a revolution in remote sensing. HSI processing has increasingly become a research hotspot and has been widely used in environmental management, greening detection [3], monitoring [4], crop analysis [5], etc. In the literature, various supervised and unsupervised approaches have been proposed to classify the HSIs.

HSI contains rich spectral information and spatial information, which brings great convenience to its classification. Most of the methods used in the past few decades are unsupervised, such as K-means [6]. In the absence of prior information, unsupervised classification has enormous advantages.

Although unsupervised classification has significant benefits in speed, supervised classification can achieve higher classification accuracy. In recent years, various supervised machine learning algorithms have been successfully applied to HSI classification. Kernel-based methods [7], support vector machine (SVM) [8], random forest [9], logistic regression [10], and neural networks [11] are amongst the most popular methods.

However, for complex HSIs, machine learning methods are still confronted with challenges. For instance, based on machine learning methods, the classification accuracy depends on a large number of labeled samples. Nevertheless, the number of labeled samples is often less than the number of bands of the HSI, which tends to cause overfitting during model training, i.e., the so-called Hughes phenomenon [12]. Therefore, it is not easy to perform the task of HSI classification based on machine learning.

With the CNN showing excellent feature extraction capabilities in computer vision [13]–[15] and natural language processing [16], the HSI classification model based on CNN has sprung up. Chen *et al.* [17] proposed a CNN to tackle the classification of HSI. Unlike traditional classification methods, the CNN model takes HSI cube as input and uses the central pixel category to represent the entire sample, which provides a new way to classify the hyperspectral data and improves the classification accuracy. To extract the more in-depth HSI features, the structure of the classification model becomes more profound and more complex. To avoid gradient vanishing or gradient explosion, the residual structure [18] is introduced. Song *et al.* [19] used the residual architecture to fuse the features of different levels to extract more discriminative features. Paoletti *et al.* [20] proposed a pyramid-like residual structure, which increases the feature maps' dimension linearly and extracts more features to achieve high-precision classification. However, these models suffer from the number of training samples because they need abundant instances to achieve considerable performance. To overcome this limitation, Xu *et al.* [21] used multisource data for classification for the first time, extracted feature from different sensors in two branches, and finally fused the features of the two tunnel for classification. Zhang *et al.* [22] divided the HSI cube into smaller regions and enhanced the extraction ability of spatial information by fusing the features from different areas. In addition, the receptive fields of these models are fixed, which may not effectively extract spatial information. To alleviate this problem, Cui *et al.* [23] exploited an image-based model to capture fuller spatial information and used the dilated convolution [24] to extract features of different receptive fields. Lee and Kwon [25] also presented a deep context neural

Manuscript received March 1, 2021; revised April 25, 2021; accepted May 11, 2021. Date of publication May 27, 2021; date of current version January 5, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62071084, in part by the Fundamental Research Funds for the Central Universities under Grant 3072021CF0805, and in part by the Nature Science Foundation of Heilongjiang Province under Grant LH2020F021. *(Corresponding author: Shan Gao.)*

The authors are with the College of Information and Communication Engineering, Harbin Engineering University, Harbin 150001, China, and also with the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, Harbin Engineering University, Harbin 150001, China (e-mail: cuiying@hrbeu.edu.cn; jbxia@hrbeu.edu.cn; wangzhiteng@hrbeu.edu.cn; gaoshan08@hrbeu.edu.cn; wangliguo@hrbeu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3080394

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

network to fully capture the spatial information of HSI through convolution kernels of different scales, which stacks the spatial information of various receptive fields to the corresponding channels, thereby extracting rich spatial information. Other CNN-based models also can be found in [26]–[29]. Although a lot of work has shown that the CNN-based model can achieve excellent classification results, its time consumption and complexity are enormous. Recently, several strategies can be found in the literature to address this problem. For instance, Wang *et al.* [30] used the dropout layer [24] and PReLU [31] to improve the training speed. Xu *et al.* [32] reduced the time consumption of the model by integrating the process of feature extraction and classifier training. Roy *et al.* [33] proposed an end-to-end spectral–spatial squeeze-and-excitation residual bag-of-feature (S3EResBoF) framework to decrease computation time while maintaining classification accuracy. Moreover, other relevant approaches were also introduced by Zhang *et al.* [34] and Jia *et al.* [35], where they used transfer learning strategy and spatial–spectral Schroedinger eigenmaps to alleviate the overfitting phenomenon caused by limited available training samples and massive model parameters.

Inspired by the signal-processing mechanism [36] of humans, vision attention modules [37] are proposed in the CNN model. The attention module can allocate limited attention resources to the core area more effectively, extract essential information, and suppress the influence of the invalid regions. Following the spectral-based approach, Wang *et al.* [38] proposed a spectral attention module to weight and select different spectral of the feature maps, which greatly alleviated the influence of redundant spectral information and noise interference on classification. Wang *et al.* [39] applied the SE module [40] to suppress interband noise and band correlation. As for the spatial-based method, Sun *et al.* [41] adopted the non-local block [42] to extract the samples' spatial information fully. Recently, the spatial–spectral joint method is becoming a trend. Zhu *et al.* [43] and Haut *et al.* [44] took ResNet [18] as the foundation and embedded attention module to learn the most valuable spatial–spectral information for classification automatically. Furthermore, Li *et al.* [45] and Ma *et al.* [46] divided the model into two branches and used different attention modules [47], [48] to extract spatial information and spectral information, respectively. Although the attention module's introduction can effectively improve the model's information extraction ability, the model's amount of calculation also increases inevitably. To address this issue, Wang *et al.* [49] proposed a local cross-channel interaction strategy to reduce the complexity of SE module. Zhu *et al.* [50] leveraged a pyramid sampling module into the non-local block to reduce the similarity calculation without sacrificing the performance.

With the appearance of deep convolution neural network (DCNN) and attention mechanism, the classification accuracy of HSI has made a qualitative leap. However, the significant increase in classification accuracy has caused a sharp proliferation in the number of model parameters and calculations. Besides, the DCNN-based models can generally achieve excellent classification results, even in the case of using very few training samples. Therefore, reducing the complexity of the model is the key to the classification of HSIs.

In this article, we propose a lightweight HSI classification model based on the spatial-spectral attention mechanism. It can significantly reduce the number of parameters and calculation of the model under the premise of ensuring the accuracy of classification. Besides, based on experiments using Grad-CAM [51] to visualize feature maps, a lightweight and efficient spatial attention module is presented to extract spatial information, which can realize high-precision classification of HSI by combining with the SE module. Our innovative contributions can be summarized as follows:

1) In this article, for the first time, a visual analysis of the feature maps of the HSI model (using Grad-CAM) is carried out, and the essence of spatial information extraction is summarized, laying a foundation for the improvement of the spatial attention module.
2) Based on the visualization experiment of feature maps, a more concise and efficient spatial attention module is proposed. This module can extract the most valuable spatial information based on the core status of the central pixel of the sample in the classification of HSI while suppressing the interference of invalid pixels. Besides, the module can be embedded in any position of the model without introducing additional hyperparameters.
3) Based on MobileNetV3 [52], a lightweight HSI classification model is proposed in this article, which can reduce complexity while attaining high classification accuracy.
4) To alleviate the problem that the difficulty of each instance is different in classification, multiclass focal-loss [53] is applied to HSI classification for the first time.

The remainder of this article is organized as follows. Section II describes related work. Section III validates the proposed methods and model in detail. In section IV, a series of experiments are carried out to verify the proposed model. Finally, section V concludes the full text and puts forward some suggestions on future development directions.

II. Relative Work

In this section, we first introduce the commonly used channel attention module and spatial attention module, then describe the depthwise separable convolution used by the lightweight model.

A. Channel Attention Mechanism

The convolution output generally contains a large number of channels, and each channel has a different influence on the prediction, so it is necessary to extract more valuable channel information selectively. Hu *et al.* [40] pulled a Squeeze-and-Excitation module (SE module) to dynamically learn the importance of each channel, which consists of two parts, squeeze and excitation operations (see Fig. 1).

1) Squeeze: Global Information Embedding
   Assuming the previous stage convolution output is $U \in R^{H\times W\times C}$, where H, W, and C represent the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

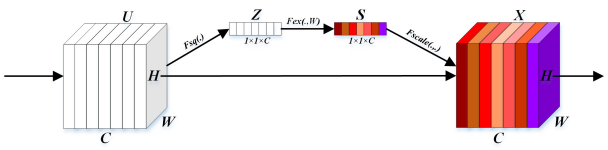


Fig. 1. Structure of SE module.

height, width, and the number of channels, respectively. SE module generates a statistical description of each channel through the global average pooling (GAP) compressed space dimension ($H \times W$), and the output result is $Z \in R^{1\times1\times C}$. Therefore, the cth channel of Z can be expressed as

$$Z_c = F_{sq}(U_c) = \frac{1}{H \times W}\sum_{m=1}^{H}\sum_{n=1}^{W} u_c(m,n). \quad (1)$$

2) Excitation: Adaptive Recalibration
To make full use of the information gathered in the squeeze operation and highlight the importance of multiple channels, in the excitation operation, only a simple threshold mechanism is used with a sigmoid function as the activation function

$$S = F_{ex}(Z, W) = \sigma(g(Z, W)) = \sigma(W_2\delta(W_1 Z)) \quad (2)$$

here, W_1 and W_2 represent the two fully connected layers, and δ means the ReLU activation function, that is, the threshold mechanism consists of two fully connected layers. The output X is obtained by rescaling U

$$X_c = F_{scale}(U_c, S_c) = s_c u_c \quad (3)$$

here $X = [x_1, x_2, \ldots, x_N]$, and $F_{scale}(X_c, \mathrm{S}_c)$ refers to channel-wise multiplication between the input U and S.

B. Spatial Attention Mechanism

In CNN, a fixed-size convolution kernel is generally used to extract spatial information, so only one local area can be processed at the same time. Only multiple cascades can capture more extensive spatial information. However, repeated local procedures would enormously increase the amount of calculation of the model and make it challenging to optimize. To solve the above problems, Wang *et al.* [42] presented a universal and straightforward spatial attention module (non-local block) to capture long-range dependencies. This module can directly calculate the dependence of any two spatial positions and can be embedded in any part of the model. Its general form can be defined as

$$y_i = \frac{1}{C(X)}\sum_{\forall j} f(X_i, X_j)\, g(X_j) \quad (4)$$

here i is the index of the output location, X means the input signal, $f(.,.)$ refers to the correlation of X at position i and position j, $g(.)$ is the representation of X at j, and $C(.)$ is a normalization function.

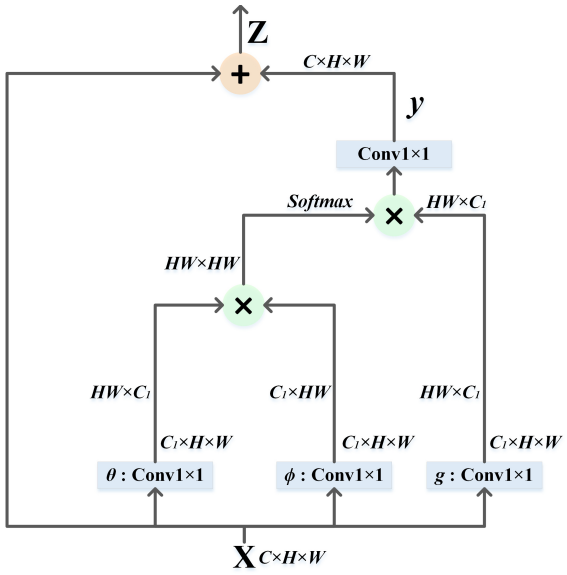


Fig. 2. Structure of non-local block, each feature map is represented by its shape. Where C is the number of channels, H and W are the height and width, respectively.

To simplify the calculation, $g(.)$ is a linear mapping, that is, $g(x_j) = W_g x_j$, W_g is the weight vector that needs to be optimized. $f(.,.)$ has several different implementation forms. Among them, Embedded Gaussian is the most frequently used one, which can be regarded as a more general form of Gaussian

$$f(x_i, x_j) = e^{\theta(x_i)^T \phi(x_j)} \quad (5)$$

where $\theta(x_i) = W_\theta x_i$ and $\phi(x_j) = W_\phi x_j$ represent two mapping forms, the normalization factor is

$$\mathcal{C}(x) = \sum_{\forall j} f(x_i, x_j). \quad (6)$$

To sum up, Fig. 2 shows the structure of the non-local block. It can be summarized as $z_i = W_z y_i + x_i$, where "$+ x_i$" is the residual connection.

C. Depthwise Separable Convolution

The depthwise separable convolution consists of two parts: depthwise convolution and pointwise convolution. Different from the standard convolution operation, depthwise convolution only processes one channel at a time, while pointwise convolution is the standard convolution with sliding window size of 1×1. Depthwise separable convolution completes the standard convolution operation in two steps, namely the cascade of depthwise convolution and pointwise convolution. Fig. 3 shows the difference between the two convolutions.

Assume that the size of the input feature map of the standard convolution is $D_i \times D_i \times N$ and the convolution output $G \in D_o \times D_o \times M$. Here, D_i and N, respectively, represent the width of the input feature map (assuming that the input feature map has the same height and width) and the number of channels. Similarly, D_o and M represent the width, height, and number of channels of the output feature map, separately. Then the parameter of standard convolution is $D_k \times D_k \times M \times N$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

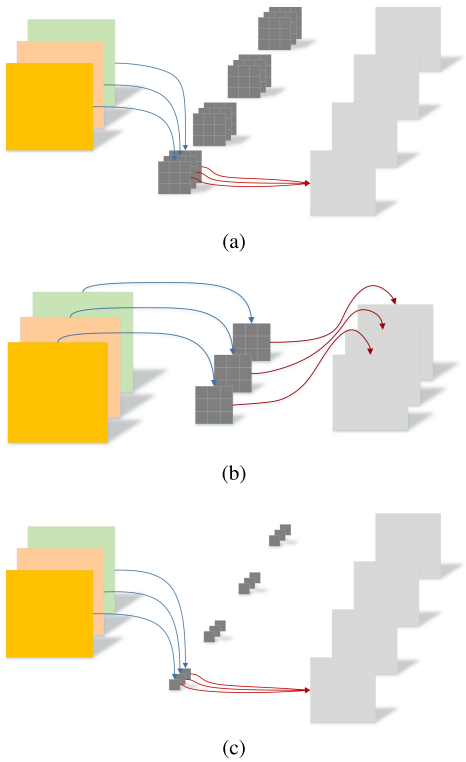


Fig. 3. Difference between standard convolution and depthwise separable convolution. (a) Standard convolution. (b) Depthwise convolution. (c) Pointwise convolution. The depthwise separable convolution is composed of depthwise convolution and pointwise convolution.

and the calculation amount is $D_k \times D_k \times M \times N \times D_o \times D_o$, where D_k is the size of the convolution kernel. In the case of the same input and output sizes, the parameter of depthwise convolution is $D_k \times D_k \times M$, and the calculation amount is $D_k \times D_k \times M \times D_o \times D_o$. The parameter of pointwise convolution is $M \times N$, and the calculation amount is $M \times N \times D_o \times D_o$, Therefore, the ratio of the parameters of the depthwise separable convolution and the standard convolution is

$$P_{D/S} = \frac{D_k \times D_k \times M + M \times N}{D_k \times D_k \times M \times N} = \frac{1}{N} + \frac{1}{D_k \times D_k} \tag{7}$$

the percentage of the calculation amount is as follows:

$$C_{D/S} = \frac{D_k \times D_k \times M \times D_o \times D_o + M \times N \times D_o \times D_o}{D_k \times D_k \times M \times N \times D_o \times D_o} = \frac{1}{N} + \frac{1}{D_k \times D_k}. \tag{8}$$

It can be seen from (7) and (8), the factorization operation drastically reduces the number of model parameters and calculations.

III. Method

In this section, we first introduce the Grad-CAM used in the feature map visualization experiment, then describe the spatial attention module and loss function related to the proposed model, and finally present the overall structure of the model.

A. Grad-CAM

In deep learning, CNN is always regarded as a black box because it is hard to describe its internal working principle with a mathematical expression. Therefore, in recent years, in the field of computer vision, the interpretability of DCNN has gradually become a trend. Whether the working mode of DCNN can be reasonably explained becomes the key to make a breakthrough in deep learning after AlexNet [54].

Grad-CAM provides a powerful tool to explain the working principle of DCNN, which uses the partial derivative as the weight factor to measure the contribution of each pixel to the prediction. Suppose that the feature map of the last convolution output of the model is F_{HW}^N, where H, W, and N represent the height, width, and the number of channels of the feature map, respectively, and the prediction category is C. The calculation process of Grad-CAM includes the following three steps:

First, the partial derivatives of the prediction probability to all the pixels on the feature are calculated

$$\frac{\partial y^C}{\partial F_{ij}^k} \tag{9}$$

where k represents the kth channel on the feature map, i and j mean the number of rows and columns, and y^C is the probability of being predicted as the category C.

Then, Average the partial derivatives of each channel on the feature map, which is similar to the GAP operation in CNN

$$\alpha_k^C = \frac{1}{H \times W} \sum_i \sum_j \frac{\partial y^C}{\partial F_{ij}^k}. \tag{10}$$

Here α_k^C represents the weight of the kth channel on the feature map, which is the contribution to the predicted probability.

Finally, each channel in the feature map is linearly superimposed according to the corresponding weight, and the final output is obtained through the activation function ReLU

$$L_{\text{Grad}-CAM}^C = \text{ReLU} \sum_k \alpha_k^C F^k \tag{11}$$

where $L_{\text{Grad}-CAM}^C$ represents an activation map, similar to a heatmap. For the record, when Grad-CAM applied to HSI classification, it is necessary to adjust the activation map to the input size, and then compare it with the real label to find the area that affects the prediction output the most.

B. Single-Point Attention

The non-local block provides an example for spatial attention module. However, it also has some limitations. First, when the input feature map is large or contains many channels, it will cause a lot of calculation. Therefore, in [50], an asymmetric spatial attention module is proposed based on spatial pyramid pooling [55] to reduce the computational complexity of the attention module. Then, when it is directly applied to HSI classification, there will be many redundant calculations, which will increase the operation time and may cause harmful effects. The input of the HSI classification model based on CNN is a patch cube and the preliminary information of each sample is only the category of the central pixel, so the central

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

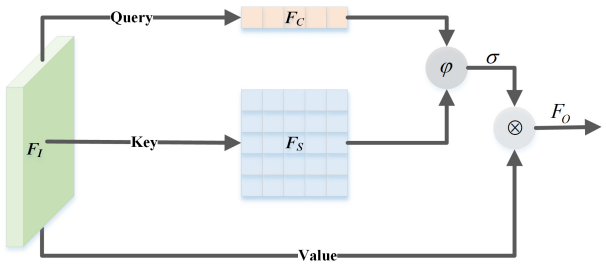


Fig. 4. Proposed SPA, Query is the central pixel, Key is the 2-D plane vector, and Value is the original input. First, the similarity between Query and Key is calculated, and then weight the corresponding position of Value according to the similarity.

pixel is used to represent the whole sample. Besides, the Grad-CAM experiments (described in section IV) show that for correctly classified samples, the corresponding visual feature maps have larger values at both the center pixel and the same class cell. In contrast, for incorrectly classified samples, there is an immense value at the interference cell. When there are many "interfering pixels" in an example, and the non-local block is directly used to extract the spatial information, it may have a negative impact and degrade the classification accuracy.

To effectively extract the spatial information of HSI, a single-point attention (SPA) block is proposed in this article (see Fig. 4). It not only emphasizes the core status of central pixel and same kind of pixel in the HSI classification but also dramatically reduces the calculation of the model. The SPA block is a computational unit which takes $F_I \in R^{H\times W\times N}$ as input. The center pixel of F_I is $F_C \in R^{1\times N}$, and the 2-D plane vector of the input is $F_S \in R^{HW\times N}$. Here H, W, and N denote the height, weight, and the number of channels of F_I. Then the SPA can be defined as

$$F_O = \sigma(\varphi(F_S, F_C)) \otimes F_I \tag{12}$$

where φ means cosine similarity, $\otimes$ denotes rescaling of input and σ is the activation function. To simplify the calculation, the cosine similarity is used to calculate the correlation between the center pixel and other pixels, and sigmoid is used to normalize the output. The SPA can also be further converted into a hard attention module by adding gate functions.

C. Multiclass Focal-Loss

In multiclass classification tasks, cross-entropy is often used to evaluate the predicted results of models. But cross-entropy is mainly used to measure the difference between two probability distributions. It is necessary to convert the output of the model to a probability distribution, which is usually accomplished with Soft max. After processing, the actual outcome is

$$p_{ic} = \text{Soft max}\left(p'_{ic}\right) = \frac{e^{p'_{ic}}}{\sum_{i=1}^{N} e^{p'_{ic}}} \tag{13}$$

where p'_{ic} is the prediction for the sample i and p_{ic} denotes the probability that the prediction of the ith instance belongs to the class c. Therefore, in the classification model, the general representation of the multiclass cross-entropy loss function is

$$\text{loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{M} y_{ic}\log(p_{ic}) \tag{14}$$

here N is the number of samples, M refers to the total number of categories, and y_{ic} means the likelihood that the label of the ith example belongs to the class c.

It can be seen from (14) that when cross-entropy is used to measure the prediction, different samples have the same proportions in the loss. However, in actual classification tasks, the difficulty of classification varies from sample to sample due to differences between examples. If the same weight is used to optimize the prediction results for each instance, the prediction results on samples that are difficult to classify will be relatively poor. In HSI classification, the number of samples of different categories is quite different, and the classification difficulty of the same kind of samples is also dissimilar. Besides, due to the influence of mixed pixels, the classification result of some instances is not ideal.

Therefore, the model needs to adaptively adjust the proportion of each instance in the loss according to the classification difficulty, to allocate more "optimized resources" to the problematic classification samples. Therefore, this article uses multiclass focal-loss to evaluate the prediction result of the model, which can be defined as

$$\text{loss}_{FL} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{M} \varphi(p_{ic}) y_{ic}\log(p_{ic}) \tag{15}$$

where $\varphi(p_{ic})$ represents the proportion of different samples in the loss function. Because p_{ic} is a randomly changing number in each prediction, the model can dynamically adjust the importance of each instance in the loss according to each prediction result.

D. Propose Lightweight Spectral–Spatial Attention Network

Considering MobileNetV3 as a lightweight feature extractor with superior performance, this article designs a lightweight spectral–spatial attention network (LSSAN) for HSI classification based on it. Since MoblieNetV3 is used for general RGB image classification, and the number of sample categories is usually much larger than the number of channels of the input picture, the feature maps' channels increase step by step as the model deepens. On the contrary, the number of channels of an HSI usually is much larger than the classes it contains. Therefore, to simplify the model's parameters, when we design the model, the number of channels of the feature maps is gradually reduced.

The proposed model consists of trans, attention module, and classification tail (see Fig. 5). Among them, trans consists of a standard convolution layer, batch normalization, and h-swish [52]. The standard convolution layer is used to ensure sufficient input information. As the core part of the model, the attention module is composed of the spatial attention module and spectral attention module, which is used to extract spatial information and spectral information of HSI.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

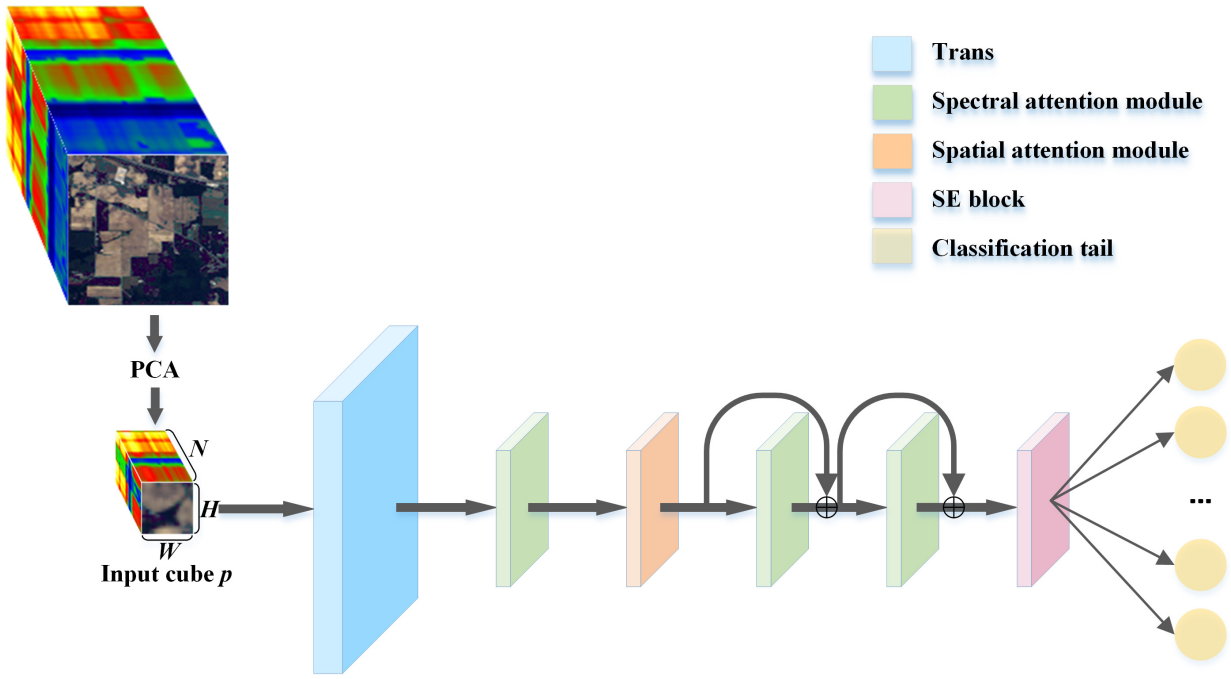


Fig. 5. Proposed lightweight spectral–spatial attention network. ⊕ means residual connection. The original HSI is reduced to the specified dimension after PCA processing, and then the input cube $p \in R^{H\times W\times N}$ is obtained by region division.

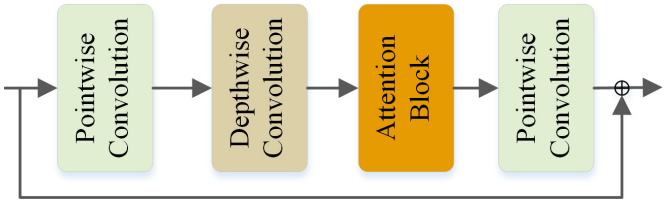


Fig. 6. Spatial attention module (or spectral attention module) structure, here, attention block refers to SPA (or SE), ⊕ means residual connection. This structure maintains a compact representation at the input and output while expanding to a high dimension in the middle. The input and output are connected with a residual connection if and only if they have the same number of channels.

The spatial attention module and spectral attention module have the same overall structure, but the "attention block" in the middle is different. Fig. 6 shows the spatial attention module (or spectral attention module) structure, which is defined by a pointwise convolution followed by depthwise convolution, attention block (SPA or SE), and pointwise convolution. The classification tail compresses the feature map into a 1-D vector by GAP, and the prediction output is then obtained through a pointwise convolution and a full connection layer. We refer to Table I for the precise layout.

IV. Experiment Results

In this section, we first introduce the hyperspectral datasets used in the experiment. Then, a battery of comparative experiments are carried out on the above datasets to verify the effectiveness of the proposed model, including comparison with some of the state-of-the-art models, and the experimental results are analyzed in detail.

TABLE I

Specification for LSSAN, Where b Is the Number of Channels Processed by PCA, c Is the Number of Categories, Exp Size Denotes Expanding Dimensions, NL Indicates the Type of Activation Function, HS Means H-Swish, RE Is ReLU, Bneck Refers to the Attention Module, and S Is Stride

Input	Operator	Exp size	Out	Attention block	NL	S
9×9×b	Conv2d, 3×3	-	b	-	HS	1
9×9×b	bneck, 3×3	128	64	SE	RE	2
5×5×64	bneck, 3×3	-	32	SPA	RE	1
5×5×32	bneck, 3×3	64	32	SE	RE	1
5×5×32	bneck, 3×3	-	32	SE	RE	1
5×5×32	SE	-	32	-	RE	-
5×5×32	GAP	-	32	-	-	-
1×1×32	Conv2d, 1×1	-	c	-	HS	1

A. Hyperspectral Datasets

The datasets used in this article are Indian Pines (IP), University of Pavia (UP), and Salinas Valley (SV). Table II shows the real label map corresponding to each dataset and the number of samples in each category. The characteristics of each dataset are described below.

1) Indian Pines, shot in northwestern Indian with AVIRIS [56] sensor in 1992. It contains 145×145 pixels, 224 spectral wavelengths ranging from 0.4 to 2.5 μm, and 16 different classes. Generally, the data of 24 invalid bands are removed.
2) University of Pavia was photographed in Pavia, northern Italy in 2002 using reflective optics spectral imaging system (ROSIS) [57]. It contains 610 × 610 pixels,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

TABLE II

NUMBER OF SAMPLES OF THE IP, UP, AND SV DATASETS

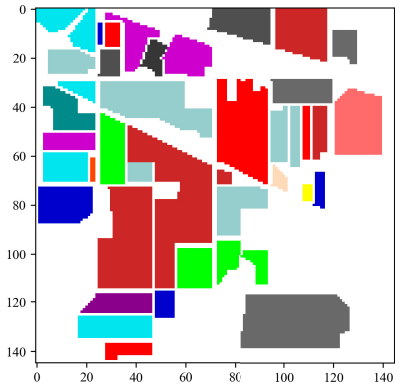
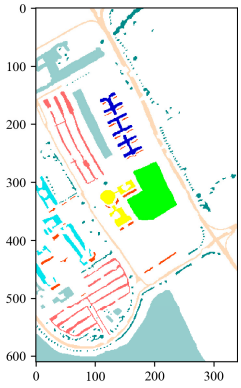
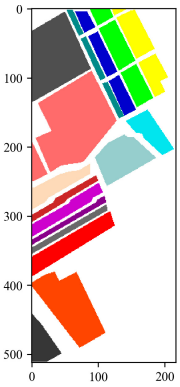
Indian Pines (IP)			University of Pavia (UP)			Salinas Valley (SV)		
								
Color	Land-cover-type	Samples	Color	Land-cover-type	Samples	Color	Land-cover-type	Samples
	Background	10766		Background	164624		Background	56975
	Alfalfa	46		Asphalt	6631		Broccoli-green-weeds-1	2009
	Corn-notill	1428		Meadows	18649		Broccoli-green-weeds-2	3726
	Corn-min	830		Gravel	2099		Fallow	1976
	Corn	237		Trees	3064		Fallow-rough-plow	1394
	Grass/Pasture	483		Painted Metal sheets	1345		Fallow-smooth	2678
	Grass/Trees	730		Bare Soil	5029		Stubble	3959
	Grass/pasture-mowed	28		Bitumen	1330		Celery	3579
	Hay-windrowed	478		Self-Blocking Bricks	3682		Grapes-untrained	11271
	Oats	20		Shadows	947		Soil-vinyard-develop	6203
	Soybeans-notill	972					Corn-senesced-green-weeds	3278
	Soybeans-min	2455					Lettuce-romaine-4wk	1068
	Soybeans-clean	593					Lettuce-romaine-5wk	1927
	Wheat	205					Lettuce-romaine-6wk	916
	Woods	1265					Lettuce-romaine-7wk	1070
	Bldg-Grass-Tree-Drives	386					Vinyard-untrained	7268
	Stone-steel towers	93					Vinyard-vertical-trellis	1807
	Total samples	21025		Total samples	207400		Total samples	111104

TABLE III

LABEL MAP AND HEAT MAP CORRESPONDING TO CORRECTLY CLASSIFIED SAMPLES

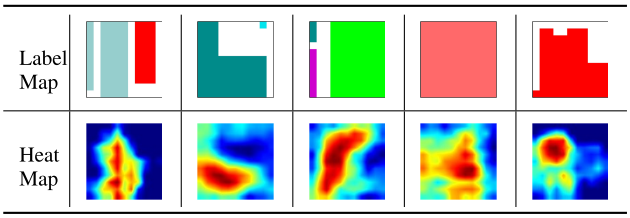
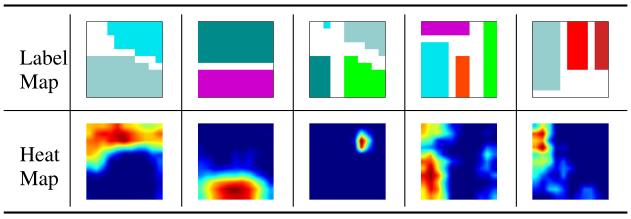


TABLE IV

LABEL MAP AND HEAT MAP CORRESPONDING TO MISCLASSIFIED SAMPLES



103 wavebands and nine kinds of ground objects, and its spatial resolution is 1.3 meter/pixel (m/p).

3) Salinas Valley, collected in California using AVIRIS sensor in 1998. It contains 512×217 pixels, 204 useful wavebands and 16 kinds of ground features, with a spatial resolution of 3.7 m/p.

B. Experimental Configuration

The network has been optimized using the Adam [58] with 200 epochs, where the learning rate decays by one-tenth on epochs 100 and 150, and the initial learning rate of IP and UP was 0.005, and 0.001 for SV. The batch size was set as 32 on IP and 128 on others. The hyperparameter γ in multiclass focal-loss was 2 on the IP dataset and 1 on the other datasets. We also used PCA [59] to reduce the dimension of all datasets. For IP, 128 bands are preserved, 16 channels are reserved for UP, and 32 bands are retained for SV. Finally, it is worth noting that the models participating in the comparison all use the parameters in the original article.

The experiments are implemented on a desktop computer with a tenth-generation Intel Core of i5-10500 processor, with 12 M of cache, the clock speed of 3.1 GHz, and six cores/twelve-way multitask processing. From the point of view

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CLASSIFICATION RESULTS FOR IP DATASET USING 5% OF THE AVAILABLE LABELED DATA

Class	DBMA	DBDA	SSRN	FDSSC	S3EResBoF	SSUN	pResNet	**Proposed**
1	89.05±0.12	98.56±0.02	96.62±0.05	**100.00±0.00**	97.23±0.04	95.00±5.46	82.73±2.31	90.48±11.17
2	93.24±0.08	96.63±0.00	94.85±0.02	**98.21±0.00**	91.88±0.02	92.48±2.23	92.28±1.67	95.82±0.49
3	94.15±0.04	95.41±0.01	95.45±0.02	94.39±0.01	**96.03±0.01**	93.87±2.69	93.84±2.12	93.26±1.67
4	95.96±0.04	92.60±0.02	93.59±0.04	**96.04±0.02**	95.45±0.02	91.77±3.69	89.34±5.73	95.44±1.73
5	94.37±0.03	**97.82±0.01**	97.61±0.01	95.74±0.01	93.99±0.02	96.51±2.55	97.39±0.89	91.17±1.46
6	96.90±0.02	98.30±0.00	99.14±0.00	98.21±0.00	94.73±0.02	95.91±2.76	98.73±0.62	**99.32±0.60**
7	**99.26±0.01**	96.42±0.05	98.89±0.02	80.00±0.40	95.09±0.08	93.33±9.77	22.96±4.92	93.08±7.05
8	99.48±0.01	**100.00±0.00**	98.59±0.01	99.81±0.00	99.76±0.00	99.38±0.53	98.94±0.87	99.49±0.72
9	80.95±0.20	**100.00±0.00**	**100.00±0.00**	80.00±0.40	**100.00±0.00**	37.9±10.21	88.42±7.74	57.78±17.78
10	91.19±0.06	98.78±0.00	96.39±0.04	**98.86±0.01**	92.08±0.02	88.14±4.39	89.23±1.87	96.07±0.49
11	93.67±0.01	97.43±0.01	96.66±0.02	97.40±0.01	93.07±0.01	96.23±1.38	96.66±1.07	**97.67±0.79**
12	86.06±0.19	**98.60±0.01**	95.90±0.01	97.47±0.01	92.94±0.04	91.56±1.72	84.83±4.92	94.88±2.34
13	99.47±0.00	99.52±0.01	99.31±0.01	**100.00±0.00**	98.70±0.01	99.08±0.75	95.90±5.45	99.24±1.51
14	97.28±0.01	97.80±0.00	97.88±0.00	**99.49±0.00**	97.27±0.01	97.24±2.79	96.74±1.34	98.82±0.84
15	90.79±0.01	94.73±0.02	93.22±0.03	98.62±.0.01	95.69±0.02	**98.70±0.53**	91.55±1.13	95.52±1.69
16	94.05±0.05	92.05±0.03	94.49±0.01	97.17±0.02	90.17±0.08	91.91±7.63	**98.18±0.91**	95.06±2.28
OA	93.21±0.03	97.33±0.00	96.49±0.01	**97.80±0.00**	94.24±0.01	94.62±0.57	94.08±0.50	96.51±0.45
AA	93.49±0.03	**97.16±0.01**	96.78±0.01	95.71±0.05	95.25±0.01	91.19±0.93	88.61±0.97	93.30±1.48
Kappa	92.26±0.04	96.95±0.00	95.99±0.01	**97.49±0.00**	93.43±0.01	93.86±0.65	93.25±0.57	96.02±0.51
Params/M	0.61	0.38	0.36	1.23	0.25	1.22	1.13	**0.20**
FLOPs/M	242.57	105.75	95.01	174.32	**1.22**	2.60	29.76	13.94
Train time/S	185.60	329.05	207.54	524.81	176.62	**35.76**	58.59	67.07
Test time/S	4.05	3.78	2.70	2.77	2.99	**0.47**	1.18	0.80

TABLE VI
CLASSIFICATION RESULTS FOR UP DATASET USING 1% OF THE AVAILABLE LABELED DATA

Class	DBMA	DBDA	SSRN	FDSSC	S3EResBoF	SSUN	pResNet	**Proposed**
1	92.43±0.03	96.61±0.01	97.21±0.03	**98.90±0.00**	94.88±0.03	92.73±1.36	94.02±2.92	96.97±1.08
2	94.85±0.02	98.75±0.00	97.67±0.01	98.18±0.01	98.70±0.01	98.73±0.85	**99.77±0.22**	99.75±0.16
3	77.52±0.13	90.93±0.12	93.29±0.03	**94.69±0.09**	91.13±0.06	69.24±7.37	72.67±1.34	75.40±3.75
4	96.17±0.04	98.00±0.01	**99.96±0.00**	98.87±0.01	94.49±0.03	96.61±1.63	91.34±2.18	96.40±0.94
5	94.22±0.08	99.77±0.00	**99.98±0.00**	99.82±0.00	89.75±0.02	99.46±0.30	99.80±0.26	99.94±0.09
6	97.61±0.01	**99.86±0.00**	99.05±0.01	99.85±0.00	99.53±0.01	92.87±2.08	95.06±2.40	99.26±0.48
7	84.64±0.19	86.13±0.27	**99.88±0.01**	99.86±0.00	96.29±0.03	81.68±3.37	74.59±2.36	97.75±1.34
8	93.19±0.03	96.29±0.02	92.93±0.03	**96.71±0.01**	83.92±0.04	94.74±0.48	90.24±2.08	85.47±3.20
9	94.58±0.05	97.53±0.01	98.80±0.02	96.47±0.01	91.31±0.06	97.59±1.56	**99.64±0.15**	97.31±1.33
OA	92.75±0.02	96.50±0.03	97.39±0.01	**98.22±0.01**	95.62±0.01	94.64±0.76	94.79±0.51	96.48±0.33
AA	91.69±0.04	95.99±0.03	97.53±0.01	**98.15±0.01**	93.31±0.02	91.52±1.23	90.79±0.51	94.25±0.65
Kappa	90.30±0.03	95.39±0.03	96.52±0.01	**97.63±0.01**	94.17±0.02	92.87±1.01	93.06±0.68	95.34±0.44
Params/M	0.32	0.20	0.22	0.65	0.22	1.20	1.11	**0.04**
FLOPs/M	144.98	67.83	48.60	90.19	1.22	2.51	28.42	**1.00**
Train time/S	57.06	168.24	111.07	170.95	110.13	**31.40**	47.59	31.66
Test time/S	10.21	9.53	6.99	6.98	6.77	**1.67**	4.43	2.35

of memory, it is equipped with 16 GB of DDR4 RAM, with a serial speed of 2666 MHz, and an NVIDIA GeForce RTX 2070 super graphical processing unit (GPU) with 8 GB RAM. The operating system is Ubuntu 16.04. To efficiently implement the proposed method, our models have parallelized on the available GPU using PyTorch.

C. Results and Discussion

To verify the classification result of the proposed LSSAN on the hyperspectral dataset, the following four experiments are carried out in this article.

1) Explore how to extract spatial information from HSI. Grad-CAM was used to visualize the feature maps of the HSI classification model (pResNet [20]) and the samples were took from IP dataset. Five correctly classified samples and five incorrectly classified instances were randomly selected for analysis. These five samples belong to classes 2, 4, 6, 8, and 10, respectively. In the incorrectly classified instances, the ninth class was selected as an alternative because the eighth category is all classified correctly.
2) Compare the performance between the proposed model and the seven best models currently available, namely DBMA [46], DBDA [45], SSRN [27], FDSSC [30], S3EResBoF [33], SSUN [32], and pResNet [20]. We tested on all three datasets mentioned in Table II. For IP datasets, 5% is randomly selected as the training set, 5% as the validation set, and the rest is used as the test. For UP and SV datasets, 1% is randomly chosen as the training set, 1% as the validation set, and the rest as the test set.
3) Verify the effectiveness of the proposed spatial attention module. There are three experimental conditions:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CLASSIFICATION RESULTS FOR SV DATASET USING 1% OF THE AVAILABLE LABELED DATA

Class	DBMA	DBDA	SSRN	FDSSC	S3EResBoF	SSUN	pResNet	**Proposed**
1	**100.00±0.00**	99.93±0.00	99.09±0.01	99.94±0.00	99.98±0.00	98.48±0.99	99.54±0.65	**100.00±0.00**
2	99.69±0.00	99.84±0.00	96.55±0.07	**100.00±0.00**	99.97±0.00	98.90±1.03	99.97±0.06	99.96±0.04
3	97.99±0.02	99.39±0.01	**100.00±0.00**	**100.00±0.00**	99.92±0.00	96.24±1.67	**100.00±0.00**	**100.00±0.00**
4	99.10±0.02	98.08±0.01	99.55±0.00	**99.87±0.00**	97.49±0.02	99.08±1.60	99.00±0.81	99.43±0.50
5	91.97±0.14	**99.63±0.00**	98.08±0.01	98.21±0.01	99.46±0.00	98.50±0.52	98.93±0.53	98.52±0.55
6	99.98±0.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.89±0.00	99.90±0.07	99.95±0.06	99.96±0.03
7	99.87±0.00	99.97±0.00	99.97±0.00	**100.00±0.00**	99.87±0.00	99.58±0.42	**100.00±0.00**	99.83±0.03
8	94.75±0.04	94.48±0.02	88.88±0.13	**98.73±0.01**	97.34±0.02	93.81±3.15	91.16±1.02	93.45±1.65
9	98.87±0.01	99..78±0.00	99.90±0.00	**100.00±0.00**	**100.00±0.00**	99.63±0.09	**100.00±0.00**	99.99±0.01
10	96.48±0.02	97.93±0.01	96.94±0.01	97.64±0.01	92.68±0.14	96.52±1.82	**98.56±0.62**	95.16±0.63
11	96.80±0.02	99.71±0.00	95.39±0.03	99.69±0.00	93.35±0.13	95.03±2.89	**99.89±0.13**	99.16±0.32
12	97.53±0.02	99.47±0.00	**99.99±0.00**	99.98±0.00	99.28±0.00	98.52±1.19	99.22±0.84	99.92±0.11
13	95.93±0.03	96.18±0.04	**100.00±0.00**	97.73±0.01	94.54±0.00	99.34±0.66	98.50±1.08	99.45±0.32
14	99.38±0.01	**99.89±0.00**	96.88±0.01	99.37±0.00	92.03±0.07	97.53±2.13	98.68±0.71	99.33±0.37
15	77.90±0.14	94.56±0.06	81.29±0.16	89.34±0.08	**94.89±0.03**	87.32±3.69	91.02±2.75	93.01±1.07
16	99.33±0.00	99.90±0.00	99.33±0.01	99.60±0.00	**99.97±0.00**	97.14±1.81	99.24±0.14	97.90±0.42
OA	93.21±0.04	97.63±0.01	94.42±0.02	**97.92±0.01**	97.42±0.02	96.05±0.33	96.66±0.41	97.18±0.42
AA	96.60±0.01	98.67±0.00	96.99±0.01	**98.73±0.01**	97.54±0.02	97.22±0.30	98.35±0.26	98.44±0.16
Kappa	92.48±0.04	97.36±0.01	93.79±0.02	**97.68±0.01**	97.12±0.02	95.60±0.37	96.28±0.45	96.87±0.47
Params/M	0.62	0.39	0.37	1.25	0.22	1.21	1.13	**0.05**
FLOPs/M	247.55	107.90	96.94	177.82	**1.22**	2.60	30.20	1.73
Train time/S	171.81	355.44	185.80	434.44	131.31	**39.65**	55.18	41.16
Test time/S	23.07	21.98	15.53	17.01	8.46	**2.10**	5.78	3.38

using SPA, using SE, and not using the attention module.

4) Compare the classification accuracy of each model under different training percentages. For IP datasets, 5%, 10%, 15%, 20%, and 25% are selected as training sets, and 1%, 3%, 5%, 7%, and 10% are selected as training sets for UP and SV, respectively.

To measure the experiment results effectively, three widely used quantitative indicators are used to test the result of classification: overall accuracy (OA), average accuracy (AA), and Kappa coefficient.

1) Experiment I (Visualize and Analyze Feature Maps): The experimental results are shown in Tables III and IV. It can be seen, for the correctly classified samples, that the center pixel and the pixels of the same category have the most obvious impact on the prediction results. For the incorrectly classified instances, due to their corresponding heat map has a larger value at the noncentral pixel category, which leads to the wrong classification. When there are fewer "interference pixels" in the sample, such as classes 6, 8, and 10 in the correctly classified samples. The vast majority of the cells around the center pixel and the center pixel belong to the same category, which facilitates the model to extract more abundant spatial information, and then correctly predict the category of the current sample. However, when there are too many interference pixels in the sample, such as classes 9 and 10 in the incorrectly classified sample, the spatial complexity of the sample increases, which makes it difficult for the model to make an accurate judgment.

As we all know, the extraction of spatial information is essential for the high-precision classification of HSI. At present, spatial attention module is generally used to extract more valuable spatial information. However, the existing spatial attention modules do not highlight the core status of the central pixel. As the only prior information for each sample, the central pixel should be given more attention resources.

From the above experiments and analysis, it can be seen that when designing the spatial attention module, it is necessary to emphasize the center pixel and pixels of the same class. At the same time, essential restrictions must be imposed on interference pixels.

2) Experiment II (Compare Classification Results Between Different Models Under the Same Training Set): The eight models involved in the comparison are all based on DCNN, but the first four models are more complex, while the remaining four are lighter. The experimental results are shown in Tables V–VII and some of the obtained classification maps are shown in Figs. 7–9.

Both SSRN and FDSSC apply 3-D-CNN to extract spatial–spectral information. The difference is that FDSSC incorporates a dense residual structure [60] in the model, whereas SSRN uses a standard residual form. In addition, in FDSSC, the input size is 9×9 and SSRN is 7×7. The introduction of the dense residual structure enables the information to be more effectively fused with the features extracted by the previous stage in the process of forwarding propagation, reducing the loss of information and enhancing the feature extraction ability of the model. As we all know, the classification ability of the model based on spectral information is lower than the model based on spatial–spectral information. The increase of input size makes the spatial information contained in the sample richer, which is beneficial to improve the classification accuracy of the model. Although FDSSC has a more remarkable improvement over SSRN, FDSSC's calculation is nearly twice that of SSRN, and the amount of its parameters is more than three times that of SSRN. DBDA is improved from DBMA, both of which are consistent in overall structure and are double-branch attention

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

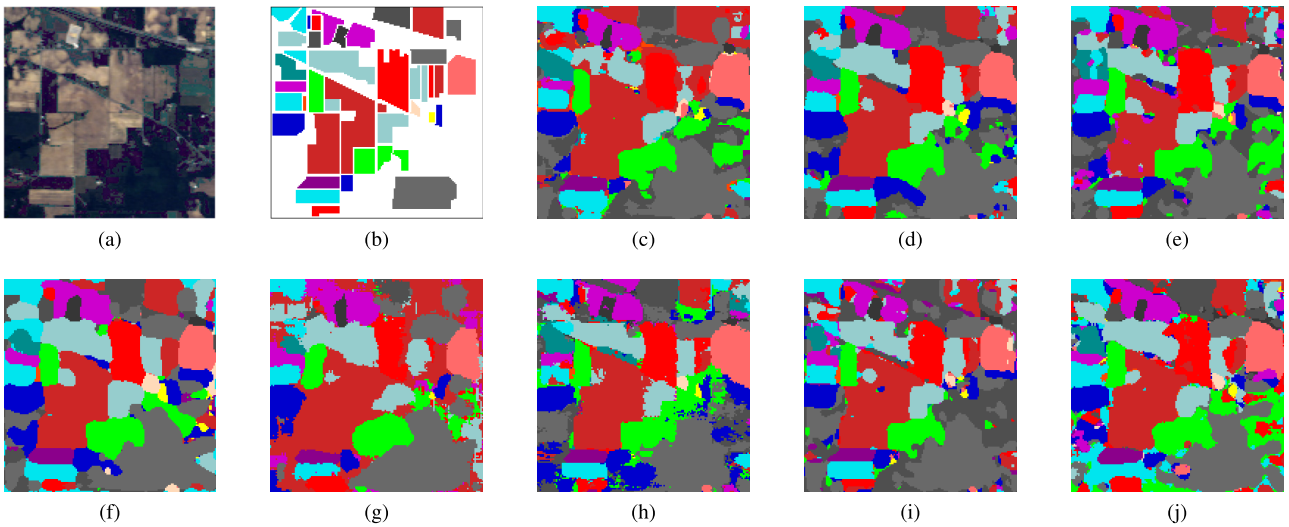


Fig. 7. Classification maps for IP dataset. (a) Simulated RGB composition of the scene. (b) Ground-truth classification map. (c) DBMA (93.21%). (d) DBDA (97.33%). (e) SSRN (96.49%). (f) FDSSC (97.80%). (g) S3EResBoF (94.246.51%). (h) SSUN (94.62%). (i) pResNet (94.08%). (j) Proposed (96.51%).

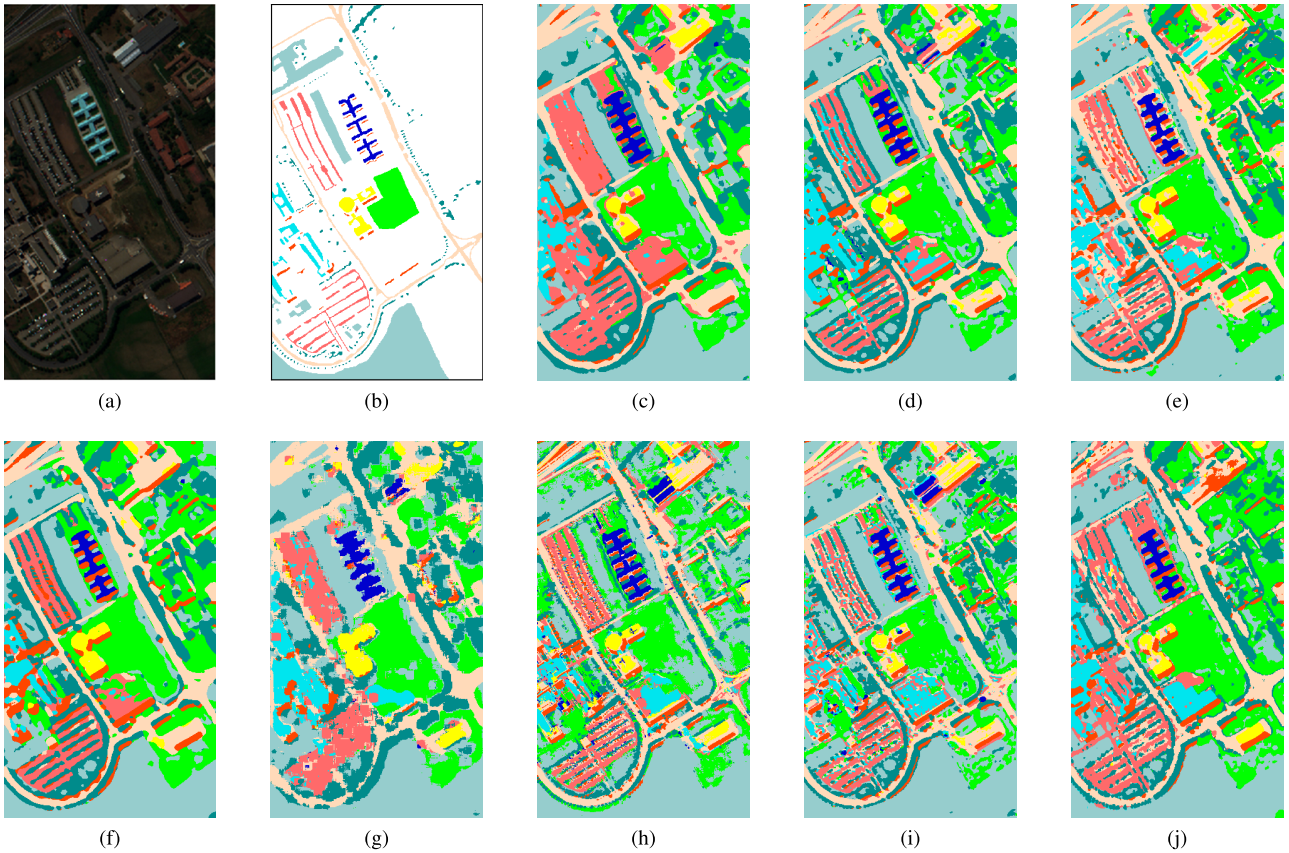


Fig. 8. Classification maps for UP dataset. (a) Simulated RGB composition of the scene. (b) Ground-truth classification map. (c) DBMA (92.75%). (d) DBDA (96.50%). (e) SSRN (97.39%). (f) FDSSC (98.22%). (g) S3EResBoF (95.62%). (h) SSUN (94.64%). (i) pResNet (94.79%). (j) Proposed (96.48%).

models. The difference is that DBMA uses CBAM [48] as the attention module, while DBDA uses PAM [47] to extract spatial information and CAM [47] to extract spectral information. In addition, DBDA adjusts the input size from 7×7 to 9×9. The increase in input size allows the model to extract more spatial information and enhances the model's discriminative ability. Similarly, compared to DBMA, the training time of DBDA has also increased significantly.

It can be seen from the above analysis that the classification accuracy of the model is generally proportional to the complexity of the model. Although DBDA and FDSSC can achieve better classification accuracy, their model complexity and time consumption are also enormous. Therefore, under the premise of ensuring classification accuracy, reducing the model's complexity becomes the key. In terms of model parameters, the proposed model is less than one-sixth of FDSSC on IP dataset, one-sixteenth on UP dataset, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

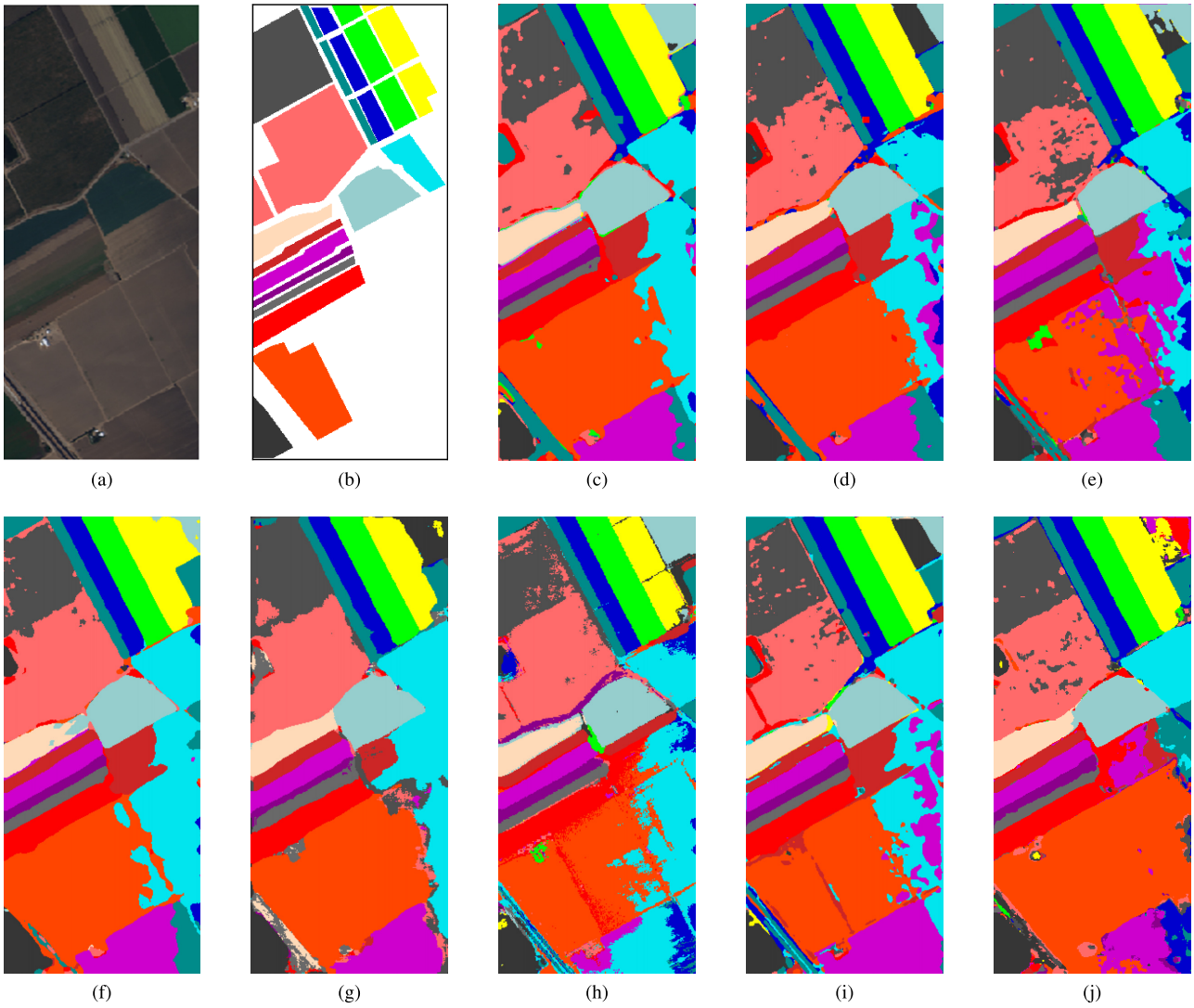


Fig. 9. Classification maps for SV dataset. (a) Simulated RGB composition of the scene. (b) Ground-truth classification map. (c) DBMA (93.21%). (d) DBDA (97.63%). (e) SSRN (94.42%). (f) FDSSC (97.92%). (g) S3EResBoF (97.42%). (h) SSUN (96.05%). (i) pResNet (96.66%). (j) Proposed (97.18%).

one-twenty-fifth on SV dataset. As far as the amount of calculation, the proposed model is less than one-twelfth of FDSSC on the IP dataset, one-ninetieth on UP dataset, and one-100th on SV dataset. However, for the IP dataset, LSSAN is no more than 1.5% lower than FDSSC in OA and Kappa. Due to the poor classification result of the ninth category, LSSAN is nearly 2.5% lower than FDSSC in AA. On the UP dataset, because the number of model parameters and the amount of calculation is significantly reduced, there exists a relatively large gap, but it is still close to DBDA. Compared with DBMA, which has a massive amount of computation, LSSAN not only dramatically reduces the complexity but also improves the accuracy of classification. For the SV dataset, there is less than 1% gap in three metrics between LSSAN and FDSSC, and even exceeds SSRN. Besides, compared with the three lighter models, the proposed model achieves better classification results on all three datasets on the whole. From the model's complexity and time consumption, LSSAN only has a large gap with SSUN on IP datasets but has a considerable performance improvement. This experiment fully demonstrates that the proposed model can simplify the model while maintaining an excellent feature extraction ability.

3) Experiment III (Verify the Effectiveness of SPA): The results show that, as shown in Fig. 10, the SE block is superior to the nonattention mechanism in classification as a whole, and the SPA block is better than the SE block. For OA and kappa, SPA's embedding has a significant improvement in UP and SV, but not in IP. For AA, there is a considerable improvement in each dataset. The experimental results show that the attention mechanism's use is more effective when the number of samples is severely limited, conducive to extracting more in-depth features. The experiment result indicates that spatial information extraction is essential in HSI classification and the spatial attention mechanism proposed by us is better than the SE block.

4) Experiment IV (Compare the Classification Result of Each Model Under Different Training Set Ratios): As we all know, the classification model based on the CNN depends heavily on the number of training sets, that is, the classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

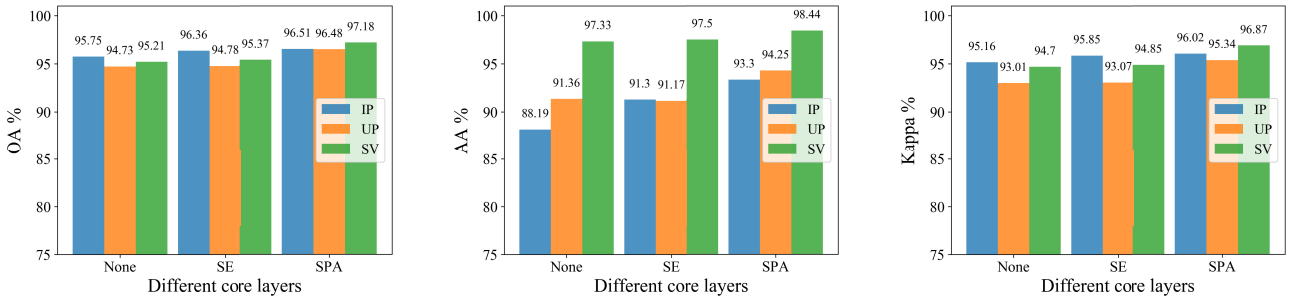


Fig. 10. Classification results on three datasets under three different structures. (a) OA. (b) AA. (c) Kappa.

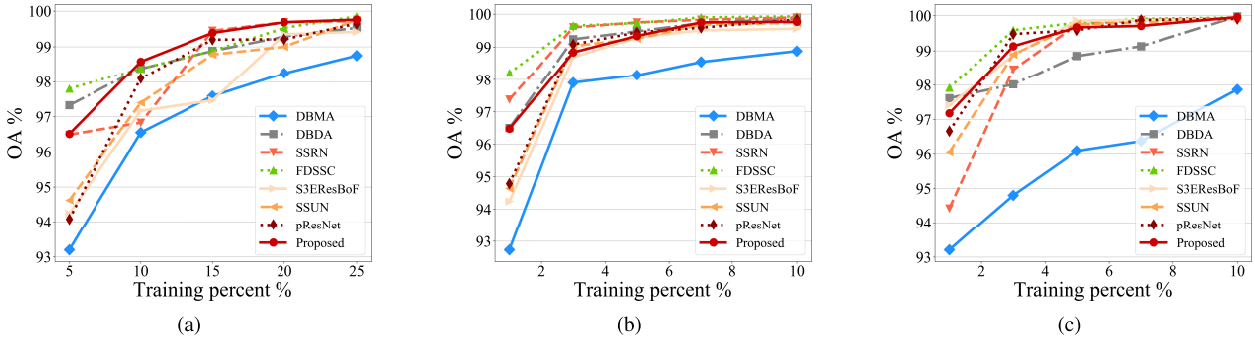


Fig. 11. Classification results of each model under different training set ratios on the three datasets. (a) IP. (b) UP. (c) SV.

accuracy of the model is proportional to the number of samples. The experimental results are shown in Fig. 11. It can be seen from the classification results of all models on each dataset that the classification accuracy of each model improves with the increase of the training set and tends to be stable gradually. When the training set on the IP dataset exceeds 10%, or the training set on the UP and SV dataset surpasses 5%, the accuracy of LSSAN is basically consistent with the best model at present.

V. Conclusion

In this article, we proposed a lightweight HSI classification model based on the spatial–spectral attention mechanism, which uses depthwise separable convolution to shrink the computation and parameters of standard convolution. Besides, to lessen the problem of a mass of redundant calculations in the existing spatial attention module, we propose a more concise and efficient spatial attention module based on the visual analysis of the feature maps of the CNN-based model. This module can be embedded in any position of the model without introducing extra hyperparameter. The multiclass focal-loss is applied to solve the problem that the difficulty of classifying each sample is different.

Experiments show that, with a handful of training samples, LSSAN can significantly decrease the complexity of the model while maintaining high accuracy. With a moderate increase in training samples, it can compete with the state-of-the-art model.

As future work, we intend to study how the HSI classification model extracts spectral information and further optimize the proposed model. Besides, we plan to explore more effective feature extraction models to simplify the parameters of the model and improve the accuracy of classification.

References

[1] H. Guo, Q. Shi, B. Du, L. Zhang, D. Wang, and H. Ding, "Scene-driven multitask parallel attention network for building extraction in high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4287–4306, May 2020.

[2] Q. Shi *et al.*, "Domain adaption for fine-grained urban village extraction from satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1430–1434, Aug. 2020.

[3] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[4] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Aerial vehicle tracking by adaptive fusion of hyperspectral likelihood maps," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 39–48.

[5] X. Zhang, Y. Sun, K. Shang, L. Zhang, and S. Wang, "Crop classification based on feature band set construction and object-oriented approach using hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4117–4128, Sep. 2016.

[6] J. M. Haut, M. Paoletti, J. Plaza, and A. Plaza, "Cloud implementation of the K-means algorithm for hyperspectral image analysis," *J. Supercomput.*, vol. 73, no. 1, pp. 514–529, Jan. 2017.

[7] G. Camps-Valls and L. Bruzzone, "Kernel-based methods for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 6, pp. 1351–1362, Jun. 2005.

[8] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[9] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[10] Z. Wu, Q. Wang, A. Plaza, J. Li, L. Sun, and Z. Wei, "Real-time implementation of the sparse multinomial logistic regression for hyperspectral image classification on GPUs," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1456–1460, Jul. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

[11] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[12] D. L. Donoho *et al.*, "High-dimensional data analysis: The curses and blessings of dimensionality," *AMS Math Challenges Lect.*, vol. 1, p. 32, Aug. 2000.

[13] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: http://arxiv.org/abs/1409.1556

[14] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*. [Online]. Available: http://arxiv.org/abs/2004.10934

[15] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[16] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," Oct. 2018, *arXiv:1810.04805*. [Online]. Available: http://arxiv.org/abs/1810.04805

[17] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[18] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 630–645.

[19] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[20] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[21] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[22] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[23] X. M. Cui, K. Zheng, L. R. Gao, B. Zhang, D. Yang, and J. C. Ren, "Multiscale spatial–spectral convolutional network with image-based framework for hyperspectral imagery classification," *Remote Sens.*, vol. 11, no. 19, p. 21, 2019.

[24] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.

[25] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[26] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[27] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[28] Y. Li, H. K. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 21, 2017.

[29] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, early access, Sep. 4, 2020, doi: 10.1109/TGRS.2020.3018879.

[30] W. J. Wang, S. G. Dou, Z. M. Jiang, and L. J. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 19, 2018.

[31] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.

[32] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[33] S. K. Roy, S. Chatterjee, S. Bhattacharyya, B. B. Chaudhuri, and J. Platoš, "Lightweight spectral–spatial squeeze-and-excitation residual bag-of-features learning for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5277–5290, Aug. 2020.

[34] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[35] S. Jia *et al.*, "A lightweight convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4150–4163, May 2021.

[36] V. Mnih, N. Heess, A. Graves, and K. Kavukcuoglu, "Recurrent models of visual attention," 2014, *arXiv:1406.6247*. [Online]. Available: http://arxiv.org/abs/1406.6247

[37] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," 2014, *arXiv:1409.0473*. [Online]. Available: http://arxiv.org/abs/1409.0473

[38] J. Wang, J. Zhou, and W. Huang, "Attend in bands: Hyperspectral band weighting and selection for image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4712–4727, Dec. 2019.

[39] L. Wang, J. T. Peng, and W. W. Sun, "Spatial–spectral squeeze-and-excitation residual network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 7, p. 16, 2019.

[40] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[41] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.

[42] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[43] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[44] J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and J. Li, "Visual attention-driven hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8065–8080, Oct. 2019.

[45] R. Li, S. Y. Zheng, C. X. Duan, Y. Yang, and X. Q. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 25, 2020.

[46] W. P. Ma, Q. F. Yang, Y. Wu, W. Zhao, and X. R. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 22, 2019.

[47] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[48] S. Woo, J. Park, J.-Y. Lee, and I. So Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[49] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11534–11542.

[50] Z. Zhu, M. Xu, S. Bai, T. Huang, and X. Bai, "Asymmetric non-local neural networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 593–602.

[51] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[52] A. Howard *et al.*, "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.

[53] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[54] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[55] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[56] R. O. Green *et al.*, "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, Sep. 1998.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.

[57] B. Kunkel, F. Blechinger, R. Lutz, R. Doerffer, H. Van der Piepen, and M. Schroder, "ROSIS (reflective optics system imaging spectrometer)—A candidate instrument for polar platform missions," *Proc. SPIE*, vol. 868, pp. 134–141, Apr. 1988.
[58] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*. [Online]. Available: http://arxiv.org/abs/1412.6980
[59] J. Shlens, "A tutorial on principal component analysis," 2014, *arXiv:1404.1100*. [Online]. Available: http://arxiv.org/abs/1404.1100
[60] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.



Ying Cui was born in Harbin, Heilongjiang, China, in 1979. She received the Ph.D. degree in signal and information processing and the Ph.D. degree in computer science and technology from Harbin Engineering University, Harbin, China, in 2006 and 2009, respectively.

At the same time, she joined the Faculty in 2005; and is an Associate Professor with the School of Information and Communication Engineering, Harbin Engineering University (HEU). Her research interests include intelligent signal processing, image processing, and wireless sensor network optimization. She is also a Reviewer for many journals, including the *Journal of Wuhan University*, the *Journal of Harbin Engineering University*, the *Journal of Applied Remote Sensing*, *Acta Optica Sinica*, and *Laser and Optoelectronics Progress*.



Jinbiao Xia received the B.E. degree in electronic information science and technology from Shanxi University, Taiyuan, China, in 2019. He is pursuing the M.S. degree in information and communication engineering with Harbin Engineering University (HEU), Harbin, China.

His research interests include deep learning, computer vision, and remote-sensing image processing.



Zhiteng Wang received the B.E. degree in optoelectronic information science and engineering from Weifang University, Weifang, China, in 2018. He is pursuing the M.S. degree in information and communication engineering with Harbin Engineering University (HEU), Harbin, China.

His research directions includes image processing and measurement of blade strain and stress in aeroengine.



Shan Gao was born in Yichun, Heilongjiang, China, in 1986. He received the Ph.D. degree in electrical engineering from the Harbin Institute of Technology, Harbin, China, in 2017.

At the same time, he joined the Faculty in 2017; and is a Lecturer with the College of Information and Communication Engineering, Harbin Engineering University, Harbin. His researches interests include artificial intelligence, image processing, and detection technology.



Liguo Wang (Member, IEEE) received the M.S. degree in natural science and the Ph.D. degree in engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2012 and 2015, respectively, and the Ph.D. degree from Harbin Engineering University (HEU), Harbin, China.

At the same time, he joined the Faculty, and is a Full Professor with the School of Information and Communication Engineering.

Dr. Wang is a Senior Member of the Chinese Institute of Electronics, a Senior Member of the China Institute of Communications (CIC), a member of the Chinese Society for Optical Engineering. He is also a fellow of the International Society for Digital Earth (ISDE). He is also a member of the Chinese Society of Optical Engineering, and a Senior Committee of Photoelectric Technology of Astronautics Society. He is also a Reviewer for many journals, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and the *Journal of Electronics and Information Technology*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:50:17 UTC from IEEE Xplore. Restrictions apply.